

How Music Refers: Music Semantics and Cross-Modal Correspondences

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Abstract

Recent theoretical work in music semantics has proposed a formal framework to model how music can convey information about the extra-musical world. At the same time, a large body of empirical research on cross-modal correspondences has evidenced systematic mappings between perceptual dimensions across sensory domains. In this paper, we present several experimental audiovisual paradigms designed to show that the two efforts should be integrated, and argue that music semantics must be grounded in cross-modal correspondences while allowing for more expressive forms of meaning. We first demonstrate that pitch and loudness can be interpreted as the level of energy of a visual object, or as its distance from a viewpoint (experiment 1). Second, we provide evidence that pitch and dynamics can be used to pick specific objects as referents (experiment 2). Third, by evidencing preferences for persistent timbre/object associations, we show that timbre can be associated with objects on the fly in the absence of any salient cross-modal correspondence, a prediction that cross-modal correspondences alone cannot make without a richer music semantic model (experiment 3). Finally, we show that these associations persist and can be reused to infer the behavior of a specific object, even in the absence of visual cues (experiment 4). Taken together, a parsimonious interpretation of these results is that music can refer to objects of the extra-musical world and simultaneously convey some properties of these objects.

1 Cross-modal correspondences and music semantics

Cross-modal correspondences (CMCs) are systematic associations between perceptual dimensions such as timbre and color brightness (Reymore & Lindsey, 2025), pitch and spatial height or size (Eitan, 2013; Eitan & Timmers, 2010), or loudness and distance (Kolarik et al., 2016; Mershon & Bowers, 1979; Neuhoﬀ, 2016, 2018). Such associations can be dynamic: intensifying (‘looming’) sounds are perceived as faster than receding ones (Neuhoﬀ, 2016,

2018) and rising intensity can bias perceived pitch upward (the “Doppler illusion”; Neuhoﬀ and McBeath (1996)). They also interact: combined changes in pitch and loudness were shown to shape motion and energy judgments (Eitan & Granot, 2006; Granot & Eitan, 2011; King & Prime, 2019) and to reduce estimated time-to-contact of visual objects (King & Prime, 2019) as well as biasing visual perception toward closer, larger, or more salient objects suggesting that such changes can be perceived as signaling approach or urgency (Changizi, 2011; Neuhoﬀ & McBeath, 1996; Spence, 2011; Sutherland et al., 2014).

CMCs thus connect properties of sounds with properties of more complete scenes (visual animations or others): some sounds may or may not be naturally associated with some scenes. This is the essence of a semantics: a relation that specifies which signals (sounds or sentences) are associated with which events in the world. Hence, it is possible to model these associations by applying some of the tools of formal semantics of natural languages, where sentences are associated with situations, those in which the sentence is true (Frege, 1892). This connection has been theorized by Schlenker (2017, 2019, 2022a, 2022b), who further developed a specific music semantics too, that is, one that specifies *how* music connects to situations in the outside world. Formally, a set of CMCs can by itself be seen as another possible music semantics. We argue that music semantics, a theoretical framework, as well as a specific parametrization of a semantic model, and CMCs, a broad range of well-described empirical facts, must be united. A full music semantics must be at least partly grounded in CMCs, since CMCs can be leveraged to identify situations a musical sequence can be associated with. At the same time, CMCs are typically constrained: they arise from fixed biases in perception or acquired conventions. Here, we attempt to show that a broader framework of music semantics allows us to understand these constraints as one part of a richer set of musical meanings.

A semantics is a function which assigns truth-values to pairs of signals and situations. For instance, the sentence “Mary runs” is true in all and only the situations, possible or imagined, in which Mary is indeed a runner. In the case of music, music semantics similarly yields truth-conditions from specific features of musical sequences, i.e. it computes constraints on what ordered sequences of eventualities an ordered sequence of sounds may denote. A musical sequence is then taken to be true of all situations, possible or imagined, that satisfy these constraints. A musical sequence instantiating a change in one sound feature (e.g., an increase in loudness) may be true of a situation in which an entity instantiates a corresponding property (e.g., approaching). Defined as such, the ‘correspondence’ relation between the sound feature and the entity’s property is very general, and in many cases, this relation reduces to a CMC (such as between loudness and distance).

However, there seem to be cases suggesting that there is more to music semantics than CMCs. Sometimes, for instance, a single musical motive is used to persistently refer to a single individual (and, crucially, not to another identical individual with the same behavior) (Schlenker, 2022a), and recent work (Di Stefano et al., 2024, 2025) shows that such associations are systematically recognized between listeners, even in nonverbal tasks, suggesting that musical sounds can function referentially, beyond simple feature-to-feature correspondences. CMCs however only constrain the compatibility with a referent at one point, and does not establish a persistent reference relation to an object. Conversely, sometimes, a contrast in a sound feature strongly evokes two distinct individuals: all things being equal, two distinct

timbres typically evoke two distinct referents.¹ Such phenomena have been accounted for by assigning to each passage, or each timbre, a given *variable* (or discourse referent): a device also used in visual semantics (Abusch, 2013) accounting for coreference across discrete pictures (i.e. explaining how one ends up identifying two objects instead of positing that one was replaced by another one). Variables can also describe coreference *across* media, typically in audiovisual animations or cartoons (such as in *The Sorcerer’s apprentice*, Schlenker, 2022a).

This suggests that music semantics can operate on two levels. One might compute the truth conditions of the passage on a note-by-note basis using minimal music semantics, possibly relying on CMCs. But one might also analyze the passage as a whole to which an individual referent is assigned. In this case, what matters is that the passage be paired with its referent first (regardless of whether it perceptually matches is), so that this passage can be later reused to refer to that referent again. A semantics that reduces to CMCs would predict that a musical sequence is associated with *any* referent instantiating the right properties, and not necessarily with the same referent over again.² By contrast, a richer music semantics that goes beyond CMCs might ensure that the right referent be singled out.

Goals and outline of experiments

This paper addresses two main questions. First, can we justify that a richer music semantics is needed by identifying semantic associations that cannot reduce to CMCs (such as reference relations between musical sequences and specific objects)? Second, what musical and sound features are involved in these associations and how do they combine?

We designed four experiments to address these questions. Experiment 1 first tests whether audiovisual animations satisfying cross-modal correspondences (between loudness or pitch and distance or energy level) are preferred compared to audiovisual animations violating the natural direction of these associations. This is directly consistent with a CMC-based approach to music semantics. Experiment 2 then shows that these associations can be leveraged to infer object identity: participants are able to integrate auditory and visual information to single out a referent in a scene containing two candidate ones. Experiment 3 then investigates whether two sounds contrasting in timbre can consistently refer to two distinct objects, irrespective of CMCs (such as between timbre and shape or color) by creating audiovisual contradictions (in which the referent of a given sound changes). It shows a clear preference for non-contradictory audiovisual scenes over contradictory ones. A pure CMC-based account cannot explain this pattern, for two reasons: the association between timbre and object is here arbitrary and created on the fly, and in a purely CMC-based approach, the property of the world that a sound feature attaches to could vary over time, jumping from one possible referent to another equally likely referent. Experiment 4 finally tests whether this timbre/object assignment survives when removing any visual cue. It shows that participants can consistently identify both the same referent and its motion even when no visual information is available, that is, independently of an actual percept, as if a story had been effectively told. Together, the results suggest that the inferential mechanisms at stake cannot simply result from observable or learned cross-modal correspondences, and justify the integration of CMCs into a more

¹This is referred to as a ‘Default principle’ in Schlenker (2022a).

²Note that this is not incompatible with the idea that some objects might be *better* candidate referents.

expressive semantics for music.

2 Experiment 1

2.1 Goals

Experiment 1 builds on both established experimental results from CMC-research on audio-visual loudness/distance and pitch/distance associations and theoretical claims from music semantics research on loudness/energy and pitch/energy associations. More specifically, it seeks to test for preferences for natural, non-arbitrary associations (e.g., decrease in loudness/visual object receding) against their exact opposite (e.g., decrease in loudness/visual object approaching). Based on the assumption that semantic match underlies preference judgments, the experiment examines which audiovisual associations are ‘acceptable’, and develops methods to detect them.³ For instance, a purely CMC-based account states that there exists a privileged perceptual mapping between an increase in loudness and a decrease in distance, that is, it predicts a preference for a congruent crescendo/object approaching association over an incongruent crescendo/object receding association. Such correspondences can also be expressed in a more general musical semantics framework (and it has been in Schlenker (2022a)). Hence, we expect that participants’ preference will reflect these CMC-based musical semantic associations.

2.2 Method

60 participants were recruited through the online platform Prolific[©] (100% based in the UK; Gender: 35 women, 19 men, 1 non-binary/self-described person; median age: 42 years, age IQR: 16.5 years). All participants first went through a few slides presenting the experiment and testing the audio level to be fixed for the rest of the experiment. All of them gave online informed consent before starting. The experiment was followed by two other experiments and an attention check was inserted right after Experiment 2, consisting in a video picked at random among all stimuli with a voice instead of a musical stimulus indicating which response to give, and to type a given word in a text box appearing on the following screen. 5/60 participants failed the attention check and were therefore removed from the data, leaving 55 participants for analysis.

For each trial, participants were presented with a pair of identical videos. One video contained a sound consistent with standard cross-modal correspondences (e.g., pitch rise and energy gain); the other video contained a mismatching sound (e.g., pitch rise and energy loss). Once both videos were played, participants were asked to click on a button under the one they preferred.⁴ The position of the congruent video on the screen was randomized. Trial order was fully randomized as well.

³For a review of methods and a justification of the use of preference tasks in linguistics, which are particularly relevant to highlight contrasts compared to interpretation judgment tasks, see Ionin and Zyzik (2014)

⁴While, as noted by a reviewer, the dichotomous choice might be seen as weakening interpretability and capturing gradient in listeners’ subjective experiences, we rely on recent findings that show that forced choice is more sensitive without inflating false positives so much for introspective tasks (Marty et al., 2020).

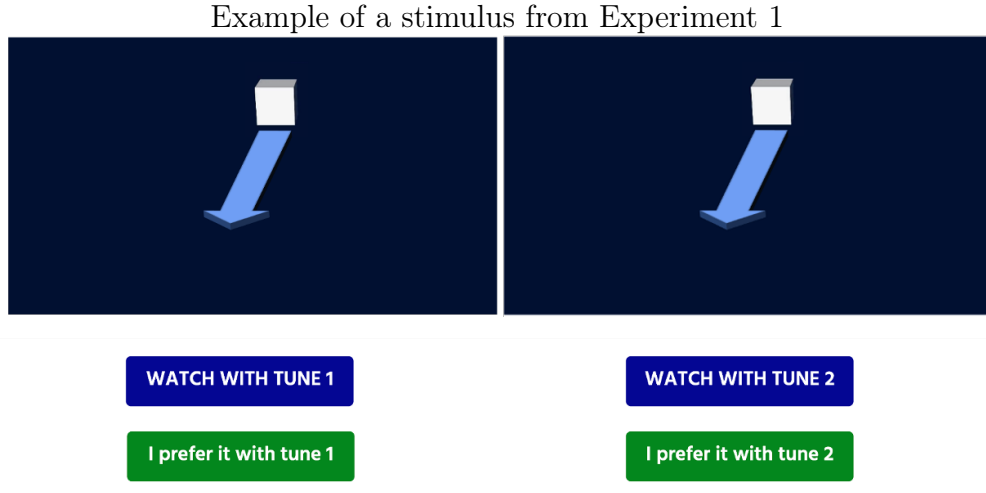


Figure 1: A congruent stimulus was presented on the left-hand side, with a cube approaching while a crescendo was played. The same video was presented on the right-hand side but with an incongruent decrescendo instead. Participants had to watch both videos so that the response buttons would appear.

2.3 Stimuli

Audiovisual animations, available [in this folder](#)⁵, were generated by combining audio stimuli generated with Garageband[®] (version 10.3.5) and visual stimuli generated with Open source software Blender[®] (version 3.4.1 2022-12-20) using the scripting interface to manipulate visual properties parametrically. The final audiovisual set resulted from crossing six musical patterns—crescendo, decrescendo, ascending and descending major and minor C scales—with four animations: object approaching, receding, gaining energy, or losing energy (energy visualized as object “shaking” with linearly increasing/decreasing amplitude). Crescendos and decrescendos were realized by linearly increasing the loudness of a repeated C tone.⁶ Tonal mode (major/minor) was varied to control for its potential interaction effect on preference judgments: as mode is known to be associated with emotional valence and intensity, it is possible that it constrains how much energy or movement is in turn associated with an object. This yielded eight congruent associations (e.g., crescendo/approach or energy gain, decrescendo/receding or energy loss; pitch rise/approach or energy gain, pitch fall/receding or energy loss) and incongruent counterparts (e.g., crescendo/receding or energy loss). Audio stimuli used the Garageband the Square Lead timbre (kept constant), chosen instead of pure tones to avoid unnaturalness reported in pilots and to minimize culture- or instrument-specific associations. Each stimulus was duplicated with either a cube or a cylinder to control for the effect of possible timbre/shape specific associations.

⁵Along with the Garageband folder containing information on loudness levels and changes, timbre, duration and tempo is accessible in the same folder.

⁶As noted by a reviewer, this resulted in a possibly confounding perceptual ‘pulsating’ effect in the crescendos and decrescendos. Experiments 3 and 4 used continuous crescendos and decrescendos instead.

			SOUND					
			Loudness		Pitch			
					Ascending		Descending	
			Crescendo	Decrescendo	Major	Minor	Major	Minor
OBJECT	Distance	Moving closer	[link]	[link]	[link]	[link]	[link]	[link]
		Moving away	[link]	[link]	[link]	[link]	[link]	[link]
	Energy	Gaining energy	[link]	[link]	[link]	[link]	[link]	[link]
		Losing energy	[link]	[link]	[link]	[link]	[link]	[link]

	Congruent
	Incongruent

Table 1: Distribution of the stimuli across conditions in Experiment 1. (All stimuli can be accessed by clicking on the [link] sign in the corresponding cell.)

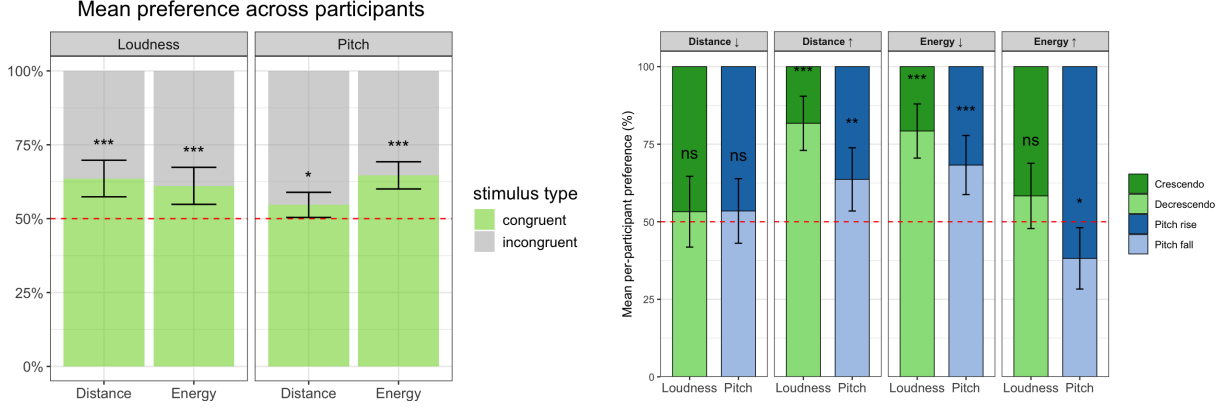
2.4 Results

The data and analysis scripts are available at [this link](#). The results in Fig. 2a show that congruent stimuli were generally preferred over incongruent ones. One-sample t -tests against 50% (chance) confirmed this for all four correspondences: loudness/distance (95% CI [0.57, 0.70], $p < .001$, Cohen’s $d = 0.42$), loudness/energy ([0.55, 0.67], $p < .001$, $d = 0.34$), pitch/energy ([0.60, 0.69], $p < .001$, $d = 0.15$) and pitch/distance ([0.50, 0.59], $p < .05$, $d = 0.43$). Due to the binary outcome of the task, non-parametric Wilcoxon signed-rank tests were also performed and yielded similar results for all four correspondences (loudness/distance: $V = 1034.5$, $p < 0.001$; loudness/energy: $V = 1039$, $p < 0.001$; pitch/distance: $V = 2413$, $p < 0.05$ and pitch/energy: $V = 5538$, $p < 0.001$).

Linear mixed-effects models corroborated these patterns. For loudness, neither object type (cube vs. cylinder) nor visual type (distance vs. energy) affected congruent choices ($\chi^2 = 0.368$, $p = .544$). For pitch, neither object type nor mode (major/minor) were significant ($\chi^2 = 0.853$, $p = .356$; $\chi^2 = 2.24$, $p = .326$), but visual type was marginally significant ($\chi^2 = 6.64$, $p = .010$), with higher endorsement for pitch/energy associations compared to pitch/distance associations.

We then break down the analysis by sound change direction (crescendo/decrecendo and pitch rise/pitch fall) and by motion direction (approaching/receding and energy gain/energy loss) in Fig. 2b, showing per-participant response compositions (means with 95% CIs for the preferred sound) for Loudness (Crescendo/Decrescendo) and Pitch (Pitch rise/Pitch fall) across motion type (Distance vs. Energy) and direction (Increase vs. Decrease). Wilcoxon tests were performed on each mean proportion against chance. The results show clear contrasts in preferences and clear interactions across motion types:

Distance. While participants preferred to associate the receding object with the Decrescendo than the Crescendo (95% CI [73.5,90], $p < 0.001$), contrasts disappeared for Distance increase (object approaching). The same asymmetry was found for pitch: while participants preferred to associate a receding object with a Pitch fall rather than a Pitch rise (95% CI [53.5,73.8], $p < 0.01$), no clear preference was found for the approaching object.



(a) Audiovisual associations between loudness and pitch, and distance and energy. (b) Breakdown of preference for specific audiovisual correspondences.

Figure 2: Results from Experiment 1. Error bars represent 95% confidence intervals for the most frequent response.

Together, these findings suggest an interaction between motion direction and sound direction in associations with distance: CMC-congruent associations are strongly preferred for receding objects, but weak or even absent for approaching ones.

Energy. Energy loss was preferably associated with both the Decrescendo (95% CI [53.5,73.8], $p < 0.001$) and the Pitch fall (95% CI [58.8,77.8], $p < 0.001$). However, while energy gain was preferably associated with Pitch rise (95% CI [51.9,71.7], $p < 0.05$), it was not significantly associated with Crescendo. Energy mappings are thus asymmetric across sound types: pitch is associated with energy change in both directions, whereas loudness is associated with energy loss but not energy gain.

Across sound types. We finally fitted a binomial mixed-effects model to the data to trial-level choices to see whether the sound preferably associated with the video depended on the property considered (loudness/pitch), on the direction of the change in that property (increase/decrease) and finally on the direction of visual movement (increase/decrease). Sound type (loudness vs pitch) had no overall main effect ($\chi^2 = 0.02, p = .888$), but visual movement type (distance vs energy; $\chi^2 = 18.92, p < .001$) and movement direction (decrease vs increase; $\chi^2 = 21.17, p < .001$) did. Crucially, the model revealed a significant interaction between sound type and motion direction $\chi^2 = 7.49, p = .006$, so that the change in distance direction was indeed stronger for loudness, while change in energy direction was stronger for pitch.

2.5 Discussion

In the CMC tradition, general principles of cross-modal associations are largely studied from the perspective of perception. In the tradition of music semantics, these general principles are largely based on inference about the source that may be producing sounds. Results here mostly replicate standard loudness/distance correspondences but also reveal loudness/energy and pitch/energy CMCs. Table 2 summarizes the main similarities and differences between CMC literature and the current findings.

Association	State of CMC Research	Current Study Findings
Loudness/Distance	Well-established loud/near correspondence; loudness as cue for proximity (Kolarik et al., 2016; Mershon & Bowers, 1979; Neuhoff, 2016, 2018)	Decrescendo preferably associated with receding object, and more so than crescendo. Unexpected asymmetry: no clear preference for crescendo or decrescendo with approaching object.
Loudness/Energy	Loudness linked to force, effort, energy (Eitan & Granot, 2006; Granot & Eitan, 2011; King & Prime, 2019)	Decrescendo preferably associated with energy decrease, and more so than crescendo. Unexpected asymmetry: no clear preference for crescendo or decrescendo with object increasing in energy.
Pitch/Distance	Mixed evidence: falling pitch linked to shrinking size (Eitan 2014) but also to increasing size / approaching objects (Fernandez-Prieto 2016; Antović 2020)	Pitch fall slightly more associated with receding than approaching object, and more so than pitch rise. No preference for pitch rise or fall with approaching object.
Pitch/Energy	Less standard; pitch sometimes linked metaphorically to vitality or tension (Antović 2020)	Symmetric preference: pitch rise preferably associated with object increasing in energy, pitch fall preferably associated with object decreasing in energy.

Table 2: Summary of associations, previous CMC findings, and results of the current study.

Interestingly, the pitch fall/receding object correspondence seems consistent with evidence that falling pitches are perceived as ‘shrinking’ (Eitan et al., 2014), since decrease in size can be perceived as receding; while more recent work suggests that pitch falls are instead associated with increases in size, arguably interpretable as objects approaching (Antović et al., 2020; Fernández-Prieto et al., 2015). A different pattern appears for loudness, which was significantly associated with distance and energy changes in one direction only (namely increase in distance and decrease in energy), and more strongly than pitch for distance.

It is possible that these asymmetries stem from ecological grounds. Loudness might arguably serve as a distance cue more often than pitch because changes in distance trigger salient loudness changes and more subtle pitch variations for sound-producing objects. Yet, this does not explain why the object approaching was not preferably associated with any change on either dimension.

A first simple explanation could be that the task involved a preference judgment, which might only superficially correlate with the perception of audiovisual fit: inconsistent associations might be perceived as incongruent and still preferred. However, should this be the main explanation, we would expect this to happen for all conditions. A second possible explanation is that both features asymmetrically map onto distance dimensions, i.e. they track receding objects better than approaching ones. This would be partly consistent with the literature on dynamic CMCs: Eitan and Granot (2006) for instance reported that “while diminuendi are strongly associated with a descending spatial motion, crescendi are not significantly associated with ascents.” A third possible explanation could be that the standard loud/near CMC was overridden by a more sophisticated inference on the approaching cube, to which

one could assign agentivity and intentions so that the cube would be expected to lower its volume when approaching the observer.

Main limitations. Experiment 1 arguably confounded distance with other co-varying visual properties, namely object size and spatial height. First, objects approaching were perceived as naturally increasing in size. Second, the bird’s eye view on the scene resulted in approaching objects being perceived at lower vertical positions in space. The pitch rise/growth association (Eitan et al., 2014) might have interacted with the expected pitch rise/downward movement association, and the crescendo/spatial descend association (Eitan & Granot, 2006) might have interacted with the expected crescendo/approaching object association.⁷ Importantly, the lack of effect of object shape and mode (major/minor) argues against stylistic confounds.⁸

Finally, it must be noted that the notion of energy we have been using remains under-specified.⁹ While in our stimuli, the most obvious energy dimension visually accessible was arguably kinetic energy, it is possible that this visual representation is either too restrictive or ill-suited for either sound feature, especially since loudness and pitch are cognitively processed in very different ways. Further work will be needed to break down this notion of energy into different concrete physical measures (kinetic, potential, thermal, etc.) and theoretically ground the possible associations between loudness and pitch height and each of these types of energy.

Overall, the results show contrasted preference patterns for audiovisual correspondences involving changes in pitch and loudness paired with changes in distance and energy. They confirm the existence of CMCs and are consistent with how this can be and has been implemented in music semantics. The next experiment tests whether these associations can give rise to cross-reference, where participants would perceive a sound and a visual object as pointing to the same referent.

3 Experiment 2

3.1 Goals

Experiment 1 evidenced preference patterns for associations between certain sound features and certain scenes, in relation to properties of a visual scene. In this experiment, we ask whether a sound can be preferentially assigned directly one of the objects (or sub-scenes) *within* a particular scene.

For concreteness, consider an audiovisual animation containing two cubes as shown in Fig. 3. The left cube is approaching and the right cube is spinning. This scene as a whole may be compatible with sounds of increasing loudness, or of constant loudness, but for different reasons: the increasing loudness is justified in relation to the approaching cube, the constant sound in association to the regularly spinning cube.

⁷These confounds were removed from the stimuli in Experiment 4.

⁸Or, as pointed out by a reviewer, this shows that mode was hardly perceivable due to the design of the scales which did not end on the tonic.

⁹We thank a reviewer for pointed this out; see Antović (2018) and Antović et al. (2024) for further discussion on the kind of energy likely to be involved in the associations we report.

If correct, this would provide reasons to refine CMCs to be applicable to subparts of a scene, not only to overall perceptual effects that a scene as a whole may generate. This would take us one step closer to a full-blown music semantics, where relations between sounds and visual events is richer than an overall similar perceptual impression.

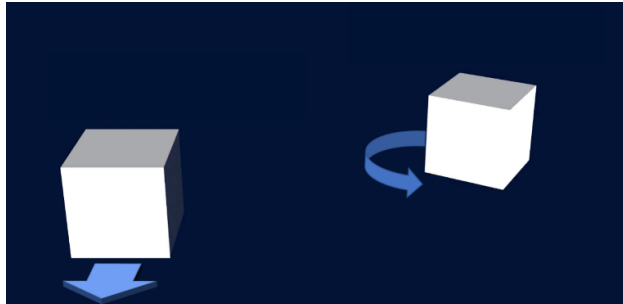


Figure 3: A scene compatible with increasing and constant loudness.

3.2 Method

Experiment 2 was administered after Experiment 1 to the same participants, randomly alternating in order with Experiment 3. A post-hoc linear mixed-effect model revealed no significant effect of order on responses from Experiment 2, be it for stimuli involving a change in loudness ($\chi^2 = 0.4192, p = 0.517$) or a change in pitch ($\chi^2 = 0.8188, p = 0.3655$).

The experiment had two within-subject conditions. In the Target condition, one sound feature covaried with the motion of a target object (e.g., decrescendo and object receding), while a second object was spinning. In the Control condition, a constant sound was repeated and expected to not favorably match any of the two objects. On each trial, participants selected one object in response to the question ‘Which object does the tune refer to?’.

3.3 Stimuli

Stimuli were audiovisual scenes that contained two identical objects, counterbalanced in shape as in Experiment 1 and randomly assigned a symmetrical left/right position. For loudness, crescendos were paired with approach or energy gain, and decrescendos with recession or energy loss. For pitch, ascending scales (major or minor) were paired with energy gain, and descending scales with energy loss¹⁰. As in Experiment 1, energy change was represented with a shaking movement, which was linearly increasing or decreasing in amplitude. In Target trials (e.g., a crescendo accompanying an approaching object while the other object was rotating at a fixed location), only one object matched the sound. In Control trials, a tone was repeated at 240 bpm with constant pitch (A4) and constant loudness, so that no object matched the sound better. In all trials, both objects moved to preclude any response strategy solely based on presence or absence of movement irrespective of whether motion matched the sound. The distribution of stimuli is summarized in Tables 3–4.

¹⁰We did not test pitch-as-distance because this mapping is absent from the semantic frameworks we rely on

		Target object			
		Moving closer	Moving away	Gaining energy	Losing energy
Sound	Crescendo	[link]		[link]	
	Decrescendo		[link]		[link]
	Constant	[link]	[link]	[link]	[link]

	Target condition
	Control condition

Table 3: Distribution of the stimuli across conditions in Experiment 2 - loudness.

		Target object	
		Gaining energy	Losing energy
Sound	Ascending pitch (major)	[link]	\
	Ascending pitch (minor)	[link]	
	Descending pitch (major)	\	[link]
	Descending pitch (minor)		[link]
	Constant	[link]	[link]

	Target condition
	Control condition

Table 4: Distribution of the stimuli across conditions in Experiment 2 - Pitch

3.4 Results

As predicted, the results in Figures 4a and 4b reveal that participants judged the sound as referring to the target object in the Target condition significantly more than in the control condition. For loudness, this was the case for both stimuli that involved a change in Distance (95% CI [0.93, 0.99] for the Target condition, [0.64, 0.80] for the Control condition; Cohen’s $d = 0.602$) and stimuli that involved a change in Energy (95% CI [0.85, 0.95] for the Target condition, [0.54, 0.71] for the Control condition; $d = 0.648$); and similarly for pitch and energy (95% CI [0.73, 0.87] for the Target condition, [0.54, 0.71] for the Control condition; $d = 0.389$).

A linear mixed-effect model revealed a significant effect of condition (control vs target) for loudness ($\chi^2 = 73.928, p < 0.001$), and similarly for pitch ($\chi^2 = 13.658, p < 0.001$). The model did not allow for testing the significance of the interaction between stimulus type (target/control) and visual interpretation (distance/energy) in the case of loudness. Qualitatively, at least, this interaction is not visible.

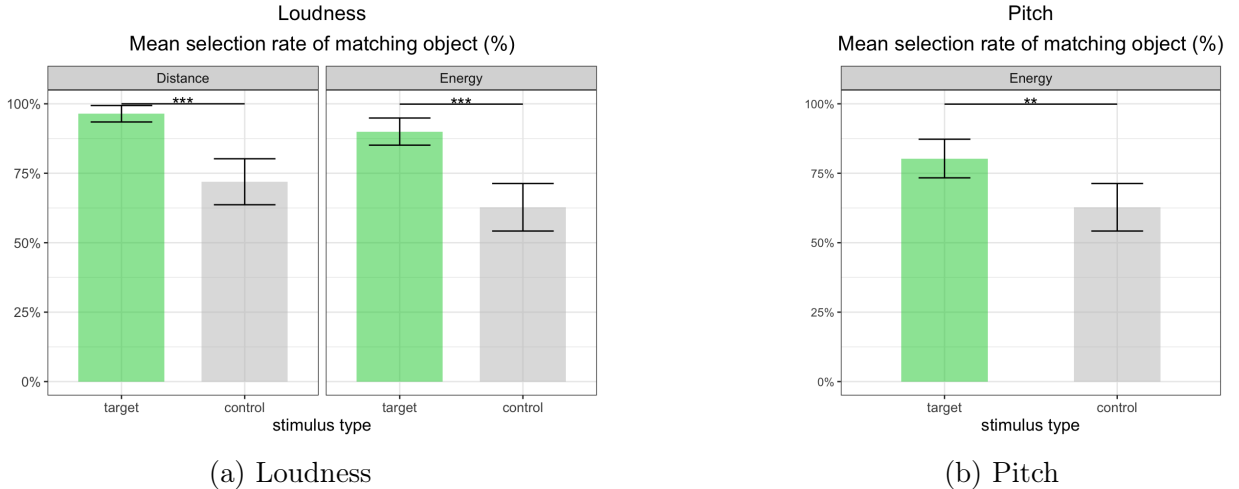


Figure 4: Results from Experiment 2

3.5 Discussion

The results suggest that a sound can be perceived as referring to an object: participants preferably associate a sound with a specific cube, based on the properties of that cube (and not just of an overall scene).

Still, the experiment faces two main limitations. First, given that the stimuli were modeled after stimuli from Experiment 1, all confounds already acknowledged still apply. A further possible confound¹¹ was that the spinning control object could be perceived as more disconnected from the sound because its period of revolution did not match the sound inter-onset intervals. However, it seems like the approaching/receding and energy gain/loss motions could just as hardly be perceived as being more consistent with the sound, since no clear period could be perceived in these motions. Second, one could argue that participants

¹¹We thank a reviewer for pointing this out.

might have produced the same choices if asked simply to “click on the object that matches the sound”, i.e. that the task was ill-suited to test for the existence of the identity inferences we are after.¹² Third, the experiment only shows that one can retrieve which object is being referred to when a single sound is heard.

Experiment 3 replaces the explicit question with an implicit preference task for audiovisual scenes involving two sounds instead of one.

4 Experiment 3

4.1 Goals

Experiment 2 showed that an association between a sound and a visual scene is not made at a global perceptual level, but in relation to subcomponents of a visual scene. In essence, there was one sound and several objects. In Experiment 3, we ask whether *two different* sounds can be used in the same global scene to distinguish *two different* objects in the scene, by testing whether two different timbres can be associated with two different objects in the same scene.

4.2 Method

Experiment 3 was administered after Experiment 1, alternating in order with Experiment 2, with the same participants. Participants were presented with pairs of minimally contrasting audiovisual scenes either consistent or inconsistent with a given timbre/object assignment, and selected the one they preferred. Video side, object side, and trial order were fully randomized.

4.3 Stimuli

The stimuli were audiovisual scenes containing two objects and either instantiating consistent timbre/object assignments (match condition) or swapped assignments (mismatch condition). In a brief pairing phase, each object was paired with one of two timbres by making each object’s forward motion co-occur with a crescendo in a distinctive timbre (cello or organ, from Garageband[©] instruments library, chosen for clear timbral contrast). This created two candidate bindings ($object_A/timbre_1$, $object_B/timbre_2$). In the test phase, one object again moved forward with a crescendo and either carried the same timbre it had been paired with earlier (Match condition, example [here](#)), or carried the other timbre (Mismatch condition, example [here](#)). Our prediction was that participants would perceive better fit and therefore prefer stimuli from the Match condition.

4.4 Results

As predicted, the results in Fig. 5 show a significant preference for stimuli from the Match condition over stimuli from the Mismatch condition (95% CI [0.59, 0.76], $d = 0.59$; $t =$

¹²As noted by a reviewer, the term ‘referring to’ used in the question can itself yield different interpretations.

4.1087, $p < 0.001$).

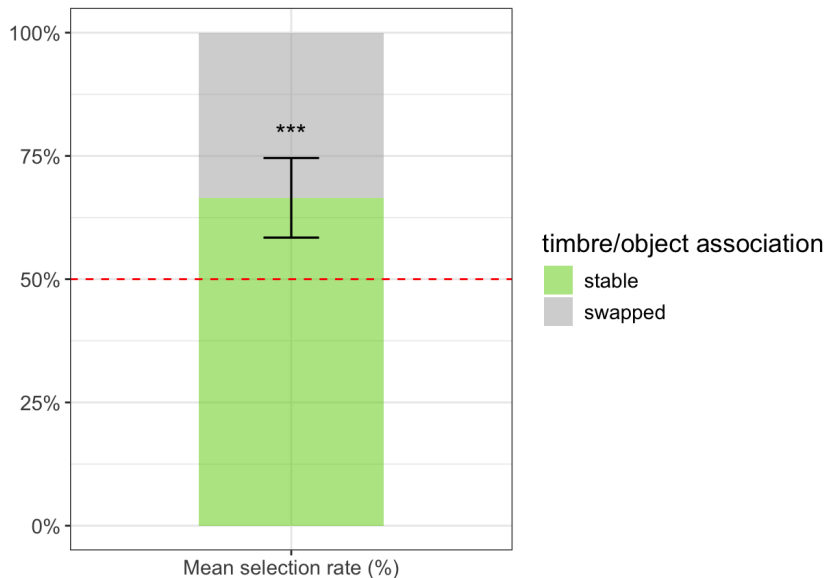


Figure 5: Results from Experiment 3. Preference for stable timbre/object associations.

4.5 Discussion

The results show a clear preference for stimuli in which timbre/object associations remained stable, consistent with the idea that each timbre was assigned one of two objects as referents.

We first note that the association between the timbres and the cubes cannot be a CMC, since no difference in the perceptual appearance of the cubes can make one timbre be preferred. Second, we note that this association is persistent across different moments of the scene, in a way that is not entailed by a pure perceptual account: the final associations are constrained by previous exposure to arbitrary associations between a timbre and a cube that go beyond perceptual alignments. There is therefore more to music semantics than CMCs: (i) it is possible for arbitrary assignments (e.g., with timbre) to be created on the fly independently of any perceptual motivation, and (ii) these assignments persist through time in audiovisual scenes.

The experiment faces some limitations, including the confounds listed for stimuli in Experiments 1 and 2¹³. The results also remain consistent with a weaker account that participants simply learn a pairing during exposure and that preference targets scenes that preserve it, without any assignment involved, so that judgments reflect a generic preference for audiovisual stability over change. In the following experiment, we abstract away from issues related to perception by fully hiding the audiovisual scene once some association has been established between timbre and object, hence placing semantic association at an inferential level beyond the perceptual level.

¹³Even though here, pitch was not manipulated, hence no interaction with vertical position could have played a role

5 Experiment 4

5.1 Goals

We ask whether participants draw inferences based on sound alone in the absence of the perceptual experience of the visual scene at all.

5.2 Method

60 new participants were recruited through Prolific[©] to take Experiment 4. Participants all declared no hearing impairment or condition affecting hearing. All participants first gave online consent, and then went through a few slides presenting the experiment and testing the audio level to be fixed for the rest of the experiment. To ensure comparability between subjects' responses, participants then went through a headphone check designed based on Woods et al. (2017) to avoid any parasitic effect on loudness perception. Participants who failed the headphone check automatically exited the experiment. An attention check was then randomly inserted in between the 5th and 11th trials, and consisted in a trial that looked like any other, except that the target sound was an AI-generated instruction for participants to choose one particular response. The experiment took about 10 minutes to complete, headphone check included. 10 participants failed either the headphone or attention check, and 4 non-native English speakers were removed from the data, leaving 46 participants for analysis (based in the UK, the US, and Canada; Gender: 20 women, 26 men; median age: 34 years, age IQR: 14 years)

Data on musical expertise was also collected along a 7-points Likert scale, following the OMSI musician-rank single index (Ollen, 2006) that has been shown to be the best predictor of both OMSI and Gold-SMI musical sophistication scores (Zhang & Schubert, 2019).

5.2.1 Task

As illustrated in Fig. 6 and at [this link](#), the task consisted in (i) watching a short audiovisual scene in which two cubes were each assigned one of two timbres, (ii) listening to a sound cue while the scene was hidden by a screen (crescendo or decrescendo with either timbre), and (iii) choosing which inference best described what happened behind the screen, among four options. Each option was consistent with none, one, or both of the two following associations: loudness/distance CMC, and timbre/object association.

Our prediction was that participants would preferably draw the target inference, i.e. the option compatible with both associations would be preferably selected: while the loudness/distance CMC should allow participants to infer motion direction, timbre should allow them to infer which object moved. If, by contrast, participants only rely on CMCs to infer what is happening, then they should randomly pick one of the two options consistent with the loudness/distance CMC, regardless of which object specifically, or they may not prefer any association at all since there is no visual simulation to form the basis of a perception-based association.

Design of a stimulus for Experiment 4

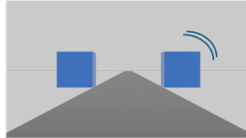
Sound	timbre ₁ paired with cube on the left	timbre ₂ paired with cube on the right	sound cue (e.g. crescendo with timbre ₁)	What happened behind the red curtain?
Visual				The object on the left moved away. The object on the right moved away. The object on the left moved closer. The object on the right moved closer.

Figure 6: Example of sound cue and circled target inference. The circle segments indicate jittering.

5.3 Stimuli

All stimuli were created using a combination of visual and audio stimuli, with audio stimuli generated through a custom Python script applied to sound samples taken from the FluidR3 soundfont library¹⁴, and visual stimuli generated with Blender. The audiovisual integration was also programmed in Blender, and the analysis script was uploaded to the OSF repository as preregistration prior to data collection.

Stimuli were audiovisual sequences that had two parts to them. During a first pairing phase, two identical cubes were each assigned one of two timbres (bassoon or violin, played with equal duration, equal perceived loudness, and at equal pitch height), by having each sound appear together with its assigned (and slightly jittering) cube. Then a red screen resembling a curtain dropped down to hide the scene. During a second, test phase, one of 4 target sounds would play for 5s: a crescendo with timbre 1, a crescendo with timbre 2, a decrescendo with timbre 1 or a decrescendo with timbre 2. Pairing sounds were matched in perceived loudness across timbres using ISO 532-1 standard so that no timbre would be perceptually heard as closer. Pairing sounds and target sounds onsets were also matched in perceived loudness to ensure each object would be imagined as departing from its seen position once occluded.

5.4 Results

The results in Fig. 7 confirm our predictions and show that participants systematically chose the target inference as the best description of what the sound cue conveyed.

We first modeled participants’ 4-alternative responses with a multinomial logistic regression including the target sound’s associated object’s side (left/right), direction (crescendo/decescendo), their interaction, and musical expertise. Likelihood-ratio tests showed a strong main effect of the cue side and of the cue direction on responses (cue side: $\chi^2(3) = 132.89, p < .001$; cue direction: $\chi^2(3) = 182.43, p < .001$), as well as a significant interaction between both cue side and direction indicating that participants were systematically most likely to choose the fully congruent answer. Musical expertise did not significantly improve model fit ($\chi^2(3) = 5.32, p = .15$).

¹⁴The FluidR3 soundfont library is available at [this repository](#) (licensed under MIT; original samples by Frank Wen).

We then fitted two binomial generalized mixed-effects models to include random intercepts for participant and video for each binary variable of interest (answer side and answer direction) and looked for possible differentiated effects of musical expertise. The first model showed a strong significant effect of cue side ($\chi^2(1) = 418.18, p < .001$), so that sounds paired with the object on the right/left were strongly associated with answers tracking right/left object. While musical expertise was also found to have a significant effect on the responses ($\chi^2(1) = 4.47, p = .035$) with higher levels of musical expertise corresponding to even stronger side bias, no interaction between musical expertise and cue side was found ($\chi^2(1) = 0.33, p = .57$). The second model showed a strong effect of cue direction (crescendo/decrescendo) on response direction (approaching/moving away) ($\chi^2(1) = 467.03, p < .001$), with no main effect of expertise ($\chi^2(1) = 0.56, p = .45$), but a significant cue direction $\times$ expertise interaction ($\chi^2(1) = 4.57, p = .032$), indicating some modulation of the directional bias by expertise.

We finally confirmed these results by comparing a binomial mixed-effects model with a fixed intercept to a chance-level model set at 0.25 and found that participants chose the target inference significantly above chance ($\chi^2(1) = 21.06, p < .001$). Musical expertise did not improve fit ($\chi^2(1) = 1.08, p = .30$), indicating that the overall advantage for the target stimulus holds across expertise levels.

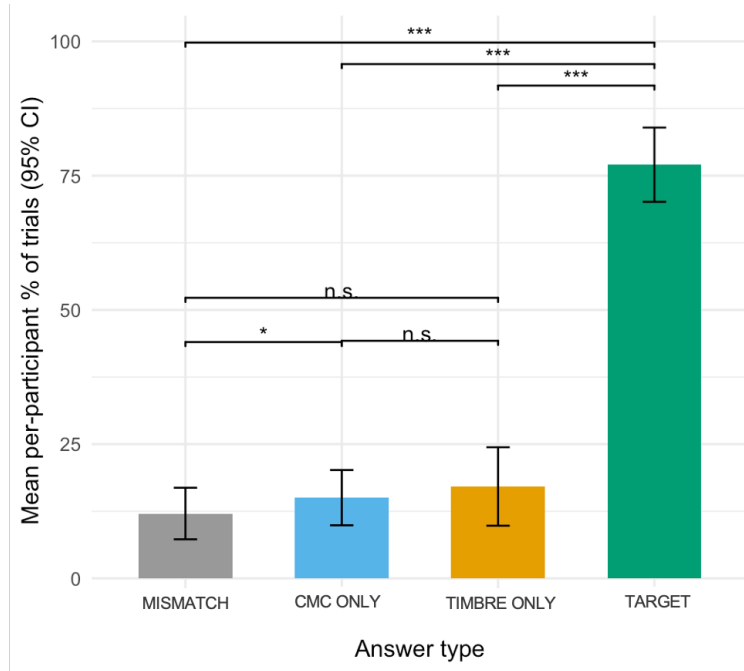


Figure 7: Results from Experiment 4: Distribution of inferential strategies from sound cues only, with 95% confidence intervals.

5.5 Discussion

The results show that participants systematically associated loudness changes to motion direction *relative to* some consistent timbre/object assignment. They are consistent with

the existence of a semantics grounded in CMCs and enriched with persistent variables that take care of selecting which of two candidate referents is singled out. Musical expertise only mildly reinforced these associations.

One could however argue that the results are partly due to task framing: the target inference might have been picked among the four forced choices because it sounded the least wrong. Even if that is true, the results are still informative: while an open-ended question might have allowed for more diverse responses, some of them unrelated to motion, the mere fact that participants would consistently agree upon which inference the sound cue licensed suggests that this inference was consistent with their semantic interpretation of the sound cue. The same issue arises in experimental tasks on language: divergent spontaneous interpretations of sentences are not incompatible with high levels of agreement upon one particular interpretation.

6 Conclusion

The four experiments aimed at identifying the mechanisms involved in music semantics and its expressivity. These include CMCs and, as we argue, are also embedded in more powerful reference mechanisms.

First, audiovisual animations are made sense of relative to cross-modal correspondences, so that associations that satisfy well-established mappings are preferred. Second, these preferences can be reused to infer which object a sound refers to when there are several similar candidate referents. Our results, however, suggest that new associations can also be created on the fly, in the absence of any prior perception-based bias (e.g., between a timbre and an object), and that they persist even when the critical visual support is removed. CMCs alone would explain neither the possible arbitrariness and persistence of associations between timbre and object, nor the association of sound with a subset of the scene. CMCs are therefore only a part of a broader music semantics, which should be set up at the featural level (see also featural interpretation for animals in Schlenker et al., 2026) to capture how sound features are combined (e.g., when each timbre is persistently assigned one of two distinct referents, while loudness should covary with distance).

Taken together, the results from all experiments support a view in which cross-modal correspondences play an important role in music semantics, creating the immediate possibility of associations and preferences among them, but demonstrate that a complete music semantics involves other mechanisms of interpretation as well, such as those involved in arbitrary and persistent reference. There has traditionally been a divide between studies focusing on perceptual mechanisms and those focusing on inferential ones, although both traditions ultimately aim to characterize some of the same mappings (described as “associations” or “meanings”, depending on the framework) using different tools and terminologies. We advocate for a fruitful and fully natural unification of the domain and for the theoretically and empirically grounded construction of a complete music semantics.

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